

DEVICE RISK ANALYSIS

Algorithms, Model Comparison, Final Model & Viva Guide

Institution	ITM Skills University
Program	B.Tech CSE • Semester V
Case Study	Case Study No. 86 – Device Risk Analysis
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This guide explains all six algorithms used in the project, their measured performance, the tuning process, why Tuned Logistic Regression was selected, and the exact viva points to remember.

1. Project at a Glance

The project is a **supervised multi-class classification** system that predicts device security risk as **Low Risk, Medium Risk, or High Risk**. The academic benchmark contains **3,012 raw records and 17 original columns**. After removing 12 duplicates and sanitizing the data, **3,000 unique records** remain.

Three engineered features are added: **Vulnerability/Firmware Ratio, Attack Surface Index, and Compliance Posture Score**. The data is split 80/20 using stratification: **2,400 training records and 600 held-out test records**.

Six classifiers are benchmarked. The strongest candidates are tuned using **5-fold Stratified GridSearchCV**. Tuned Logistic Regression with **C = 5.0** becomes the champion.

RAW	CLEAN	TEST	CHAMPION
3,012 rows	3,000 rows	600 devices	Tuned Logistic Regression

2. Model Comparison — All Algorithms

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (Tuned)	0.8900	0.8900	0.8900	0.8898	0.9756
Logistic Regression	0.8883	0.8884	0.8883	0.8883	0.9759
Support Vector Machine	0.8800	0.8802	0.8800	0.8801	0.9710
Gradient Boosting	0.8450	0.8476	0.8450	0.8452	0.9551
Random Forest	0.8017	0.8101	0.8017	0.8021	0.9265
K-Nearest Neighbors	0.7883	0.7908	0.7883	0.7868	0.9151
Decision Tree	0.7167	0.7327	0.7167	0.7178	0.8466

These values are the model-comparison metrics reported in the supplied project documentation. The tuned Logistic Regression row is the final champion evaluation on the held-out test set.

3. Explanation of Every Algorithm

Logistic Regression

A supervised classification algorithm that estimates class probabilities from the input features. In this project it handles three risk classes. It is relatively interpretable because its learned coefficients can be inspected.

Why included: It was one of the strongest baseline models and became the champion after tuning.

Viva: “Logistic Regression estimates the probability of each risk class and selects the most likely class.”

Decision Tree

A rule-based classifier that repeatedly splits the data using feature conditions, producing a tree of decisions.

Why included: It was included as an intuitive non-linear baseline, but its reported accuracy was the lowest among the six.

Viva: “A Decision Tree predicts by following a sequence of feature-based decision rules.”

K-Nearest Neighbors (KNN)

KNN classifies a new device using the labels of the most similar or nearest training examples.

Why included: It provides a similarity-based baseline and gives a useful contrast to models that learn explicit boundaries.

Viva: “KNN looks at nearby similar devices and uses their classes to make the prediction.”

Support Vector Machine (SVM)

SVM learns a decision boundary that separates classes effectively. With appropriate preprocessing it can model complex boundaries.

Why included: It was a strong benchmark and performed close to Logistic Regression.

Viva: “SVM tries to find a strong separating boundary between the risk classes.”

Random Forest

Random Forest combines many decision trees and aggregates their predictions.

Why included: It was tested as a robust tree-ensemble method, but its benchmark result was below Logistic Regression and SVM.

Viva: “Random Forest combines many trees so their aggregated prediction is more robust than a single tree.”

Gradient Boosting

Gradient Boosting builds models sequentially, with later models focusing on errors made by earlier models.

Why included: It was included as another strong ensemble approach and was also tuned, but its best validation result remained below Logistic Regression.

Viva: “Gradient Boosting builds models step by step, improving the errors of earlier models.”

4. Why the Final Model Is Called “Tuned” Logistic Regression

“Tuned” does not mean a new algorithm. It means the same Logistic Regression algorithm was tested with different hyperparameter settings to find a better configuration.

Hyperparameter: C

The project searched Logistic Regression values $C \in \{0.1, 1, 5, 10\}$. The best value was $C = 5.0$.

GridSearchCV

GridSearchCV combines a grid of candidate settings with cross-validation. The project used **5-fold stratified cross-validation** and optimized **weighted F1**. Logistic Regression reached **CV weighted F1 = 0.9084** with $C=5.0$.

5. Why Was Tuned Logistic Regression Selected?

Reason	What it means in this project
Strong baseline	Untuned Logistic Regression reported 0.8883 accuracy, higher than the other untuned models in the supplied comparison.
Best tuning result	C=5.0 gave a 5-fold cross-validation weighted F1 of 0.9084.
Strong final test	The tuned model achieved 89.00% accuracy and 0.8898 weighted F1 on the unseen 600-device test set.
Strong discrimination	Weighted one-vs-rest ROC-AUC was 0.9756.
Interpretability	Coefficients can be examined to understand model-associated feature contributions without claiming causation.
Deployment fit	The trained preprocessing + model pipeline can be serialized with Joblib and loaded by the Streamlit application.

Why not the other models?

- Decision Tree: 0.7167 accuracy — substantially below the final Logistic Regression.
- KNN: 0.7883 accuracy — below the Logistic Regression result.
- Random Forest: 0.8017 accuracy — below Logistic Regression.
- Gradient Boosting: 0.8450 accuracy — below Logistic Regression.
- SVM: 0.8800 accuracy — close, but below the final tuned Logistic Regression accuracy.

The selection is therefore evidence-based rather than a claim that Logistic Regression is universally the best algorithm.

6. Evaluation Metrics — Simple Meaning

Metric	Simple meaning	Final result
Accuracy	Percentage of all test predictions that were correct.	89.00%
Precision	When the model predicts a class, how often it is correct.	0.8900
Recall	How much of the actual class the model successfully identifies.	0.8900
F1	A balance between precision and recall.	0.8898 weighted
ROC-AUC	How well the model separates classes across scoring thresholds.	0.9756 weighted OVR

Class-wise recall

Class	Recall
Low Risk	0.93
Medium Risk	0.87
High Risk	0.88

7. How the Model Was Trained

Step	What happened
1. Load data	3,012 raw device records × 17 columns.
2. Audit	Check missing values, duplicate rows and categorical inconsistencies.
3. Clean	Remove 12 duplicates; impute missing values; standardize categories.
4. Engineer features	Create Vulnerability/Firmware Ratio, Attack Surface Index and Compliance Posture Score.
5. Split	80/20 stratified split: 2,400 training and 600 testing.
6. Preprocess	Numerical: median imputation + StandardScaler. Categorical: most-frequent imputation + OneHotEncoder(handle_unknown='ignore').
7. Train six models	Logistic Regression, Decision Tree, KNN, SVM, Random Forest, Gradient Boosting.
8. Compare	Accuracy, precision, recall, F1 and ROC-AUC.
9. Tune	5-fold Stratified GridSearchCV; weighted F1 is the optimization metric.
10. Select champion	Logistic Regression with C=5.0 and CV weighted F1=0.9084.
11. Final evaluation	Evaluate on the untouched 600-device test set: 89.00% accuracy.
12. Deploy	Save the complete pipeline with Joblib and load it from Streamlit.

8. End-to-End Flow

RAW DATA	3,012 × 17
↓	
CLEANING	12 duplicates removed → 3,000
↓	
FEATURE ENGINEERING	+ 3 domain features
↓	
TRAIN / TEST	2,400 / 600
↓	
PREPROCESSING	Impute + Scale + One-Hot Encode
↓	
6 CLASSIFIERS	LR • DT • KNN • SVM • RF • GB
↓	
GRIDSEARCHCV	5-fold stratified • weighted F1
↓	
CHAMPION	Tuned Logistic Regression • C=5.0
↓	

FINAL TEST	89.00% accuracy • 0.8898 F1 • 0.9756 ROC-AUC
↓	
STREAMLIT	Saved Joblib pipeline → prediction

9. Viva Questions & Answers

Question	Answer
Which algorithms did you use?	Six: Logistic Regression, Decision Tree, KNN, SVM, Random Forest and Gradient Boosting.
Which one did you finally use?	Tuned Logistic Regression.
Why Logistic Regression?	It was one of the strongest baselines; after tuning it achieved the best cross-validation weighted F1 among the tuned models.
What does tuned mean?	The algorithm stayed Logistic Regression; different hyperparameter settings were tested to find the best configuration.
What was the best C?	5.0.
What tuning method?	5-fold Stratified GridSearchCV using weighted F1 as the optimization metric.
What was the CV weighted F1?	0.9084.
What was the final test accuracy?	89.00% on 600 unseen test devices.
Final weighted F1?	0.8898.
Final weighted ROC-AUC?	0.9756.
Why not Random Forest?	Its reported benchmark accuracy was 0.8017, below Logistic Regression's 0.8883 baseline and the final tuned result.
What is the target?	Risk_Level.
What are the classes?	Low Risk, Medium Risk, High Risk.
Is the dataset labelled?	Yes. Each record has Risk_Level, so the task is supervised learning.
Train/test size?	2,400 / 600 after cleaning, using an 80/20 stratified split.

10. One-Minute Answer to Memorize

“My Device Risk Analysis project is a supervised multi-class classification system that predicts Low, Medium or High device risk. I benchmarked six algorithms: Logistic Regression, Decision Tree, KNN, SVM, Random Forest and Gradient Boosting. I compared them using accuracy, precision, recall, F1 and ROC-AUC. I then used 5-fold Stratified GridSearchCV to tune the strongest candidates. Logistic Regression with C=5.0 achieved a cross-validation weighted F1 of 0.9084 and was selected as the champion. On the unseen 600-device test set it achieved 89% accuracy, 0.8898 weighted F1 and 0.9756 weighted one-vs-rest ROC-AUC. I saved the complete pipeline with Joblib and connected it to the Streamlit dashboard.”

11. Final Revision Card

Topic	Answer
Problem type	Supervised multi-class classification
Target	Risk_Level
Classes	Low / Medium / High
Raw dataset	3,012 rows × 17 columns
Clean dataset	3,000 unique records
Train/Test	2,400 / 600

Topic	Answer
Algorithms	LR, DT, KNN, SVM, RF, GB
Tuning	5-fold Stratified GridSearchCV
Champion	Tuned Logistic Regression
Best C	5.0
CV weighted F1	0.9084
Test accuracy	89.00%
Weighted F1	0.8898
Weighted OVR ROC-AUC	0.9756
Deployment	Streamlit + Jolbib

Prepared from the supplied Device Risk Analysis project documentation and report. The metrics in this guide are reproduced from those project materials.